

Data Science Presentation

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Our Research Question

To what extent do technology, tourism, pharmaceutical, and energy stocks differ in terms of their price volatility and recovery dynamics during and after the COVID-19 pandemic?

Data Extraction and Cleaning

Stock Market Data

Example Stock

	▲	◀	◀	◀	◀	◀	◀
	Date	AAPL.Open	AAPL.High	AAPL.Low	AAPL.Close	AAPL.Volume	AAPL.Adjusted
1	2019-01-02	38.7225	39.7125	38.5575	39.4800	148158800	37.53882
2	2019-01-03	35.9950	36.4300	35.5000	35.5475	365248800	33.79967
3	2019-01-04	36.1325	37.1375	35.9500	37.0650	234428400	35.24256
4	2019-01-07	37.1750	37.2075	36.4750	36.9825	219111200	35.16412
5	2019-01-08	37.3900	37.9550	37.1300	37.6875	164101200	35.83445
6	2019-01-09	37.8225	38.6325	37.4075	38.3275	180396400	36.44299
7	2019-01-10	38.1250	38.4925	37.7150	38.4500	143122800	36.55946

Combined Stock List

	Apple	BioNTech	Lufthansa	Shell
2020-03-31	63.5725	58.40	6.110529	34.89
2020-04-01	60.2275	52.92	5.715150	35.31
2020-04-02	61.2325	56.91	5.692312	38.96
2020-04-03	60.3525	52.75	5.673756	36.24
2020-04-06	65.6175	52.73	6.174760	37.33
2020-04-07	64.8575	45.60	6.244700	37.26
2020-04-08	66.5175	44.95	6.217581	38.07
2020-04-09	66.9975	45.41	6.337479	37.91
2020-04-13	68.3125	41.71	NA	37.37
2020-04-14	71.7625	41.63	6.260401	37.01
2020-04-15	71.1075	38.58	5.909271	34.36
2020-04-16	71.6725	38.79	5.823629	32.60

```
1 # Merging all closing-price time series by Date (keeps all dates; missing values b
2 merged <- merge(LHA_cl, BNTX_cl, AAPL_cl, SHEL_cl, all = TRUE)
```

COVID-19 Dataset (John Hopkins University)

Data type: Daily confirmed COVID cases, continuously updated and well known for academic research and policy analysis

Filter												
Cols: << 1 - 50 >>												
	Province.State	Country.Region	Lat	Long	X1.22.20	X1.23.20	X1.24.20	X1.25.20	X1.26.20	X1.27.20	X1.28.20	X1.29
38		Cabo Verde	16.530000	-23.041000	0	0	0	0	0	0	0	0
39		Cambodia	11.550000	104.916700	0	0	0	0	0	1	1	
40		Cameroon	3.848000	11.502100	0	0	0	0	0	0	0	
41	Alberta	Canada	53.933300	-116.576500	0	0	0	0	0	0	0	
42	British Columbia	Canada	53.726700	-127.647600	0	0	0	0	0	0	1	
43	Diamond Princess	Canada	0.000000	0.000000	0	0	0	0	0	0	0	
44	Grand Princess	Canada	0.000000	0.000000	0	0	0	0	0	0	0	
45	Manitoba	Canada	53.760900	-98.813900	0	0	0	0	0	0	0	
46	New Brunswick	Canada	46.565300	-66.461900	0	0	0	0	0	0	0	
47	Newfoundland and Labrador	Canada	53.135500	-57.660400	0	0	0	0	0	0	0	
48	Northwest Territories	Canada	64.825500	-124.845700	0	0	0	0	0	0	0	
49	Nova Scotia	Canada	44.682000	-63.744300	0	0	0	0	0	0	0	
50	Nunavut	Canada	70.299800	-83.107600	0	0	0	0	0	0	0	
51	Ontario	Canada	51.253800	-85.323200	0	2	3	3	3	3	3	
52	Prince Edward Island	Canada	46.510700	-63.416800	0	0	0	0	0	0	0	
53	Quebec	Canada	52.939900	-73.549100	0	0	0	0	0	0	0	
54	Repatriated Travellers	Canada	NA	NA	0	0	0	0	0	0	0	
55	Saskatchewan	Canada	52.939900	-106.450900	0	0	0	0	0	0	0	
56	Yukon	Canada	64.282300	-135.000000	0	0	0	0	0	0	0	
57		Central African Republic	6.611100	20.939400	0	0	0	0	0	0	0	
58		Chad	15.454200	18.732200	0	0	0	0	0	0	0	

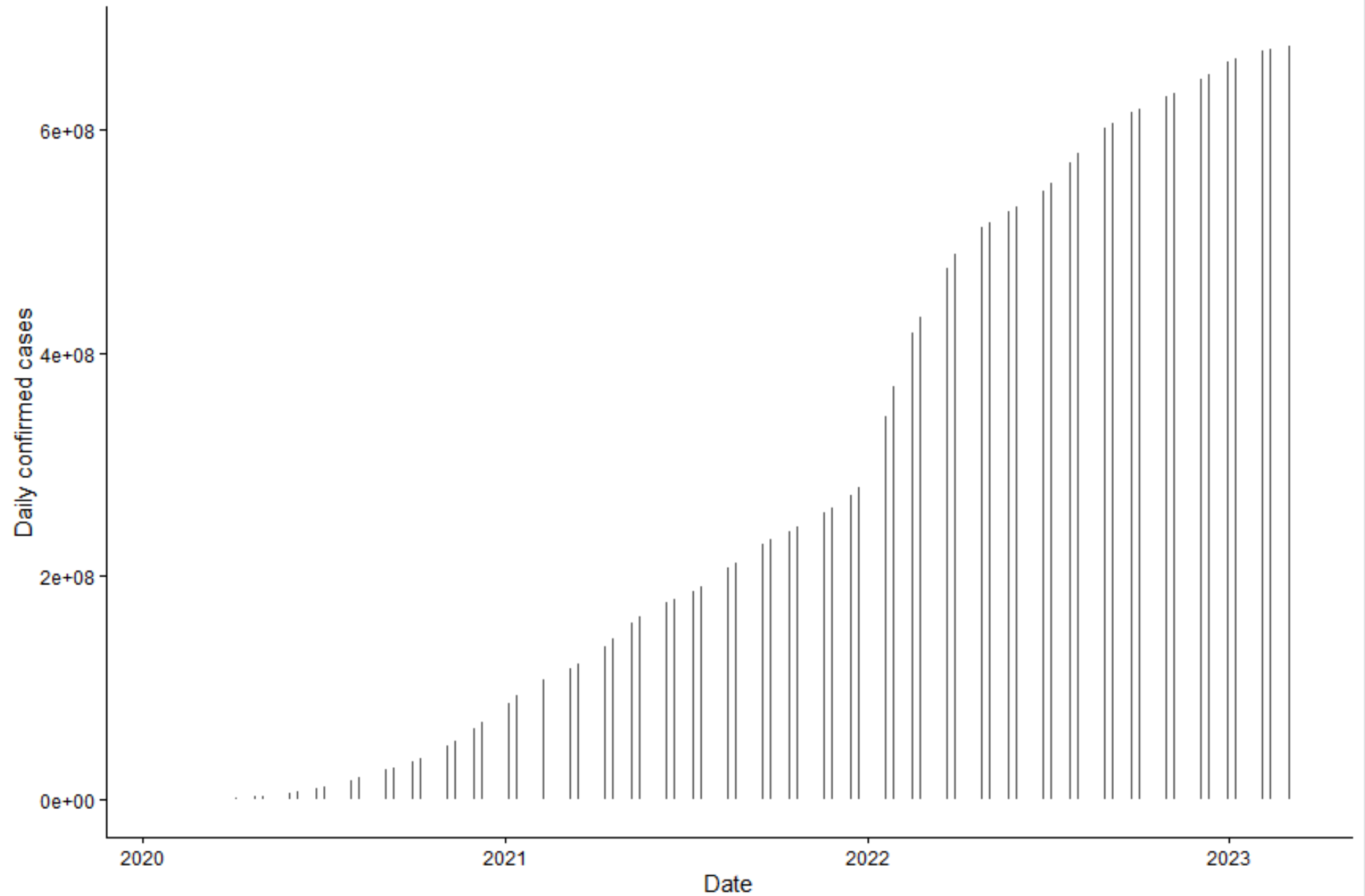
Cleaning of COVID Data

- Changing from Wide Format → long Format, so every row corresponds
- Provincial → country aggregation
- Cumulative → daily values
- Global aggregation

Final COVID Dataset: Daily global confirmed cases, daily global deaths & time index

Visualization of COVID Data

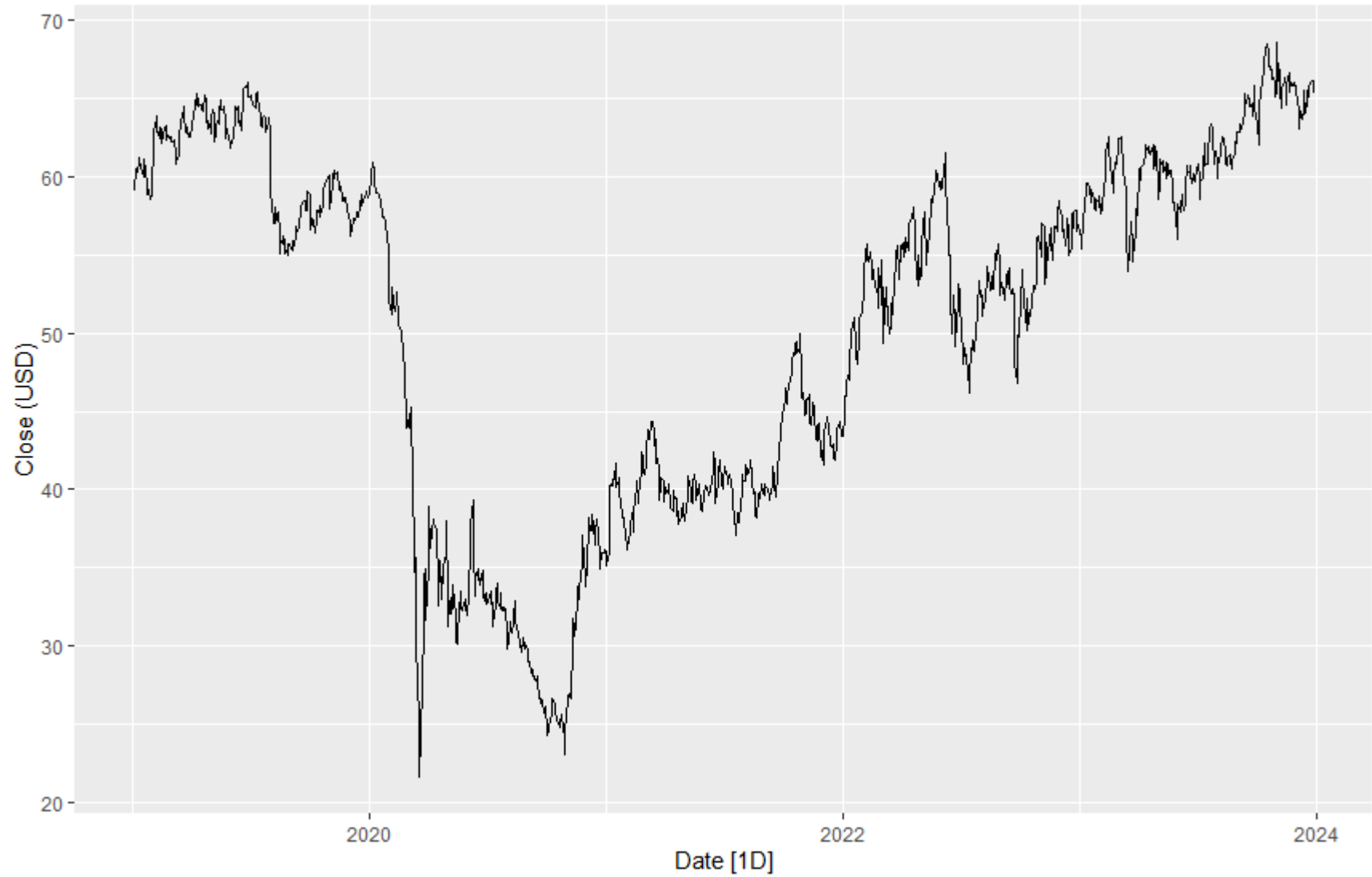
Covid-19 Global Confirmed Cases



Time Series Plots Examples: Shell and BioNtech

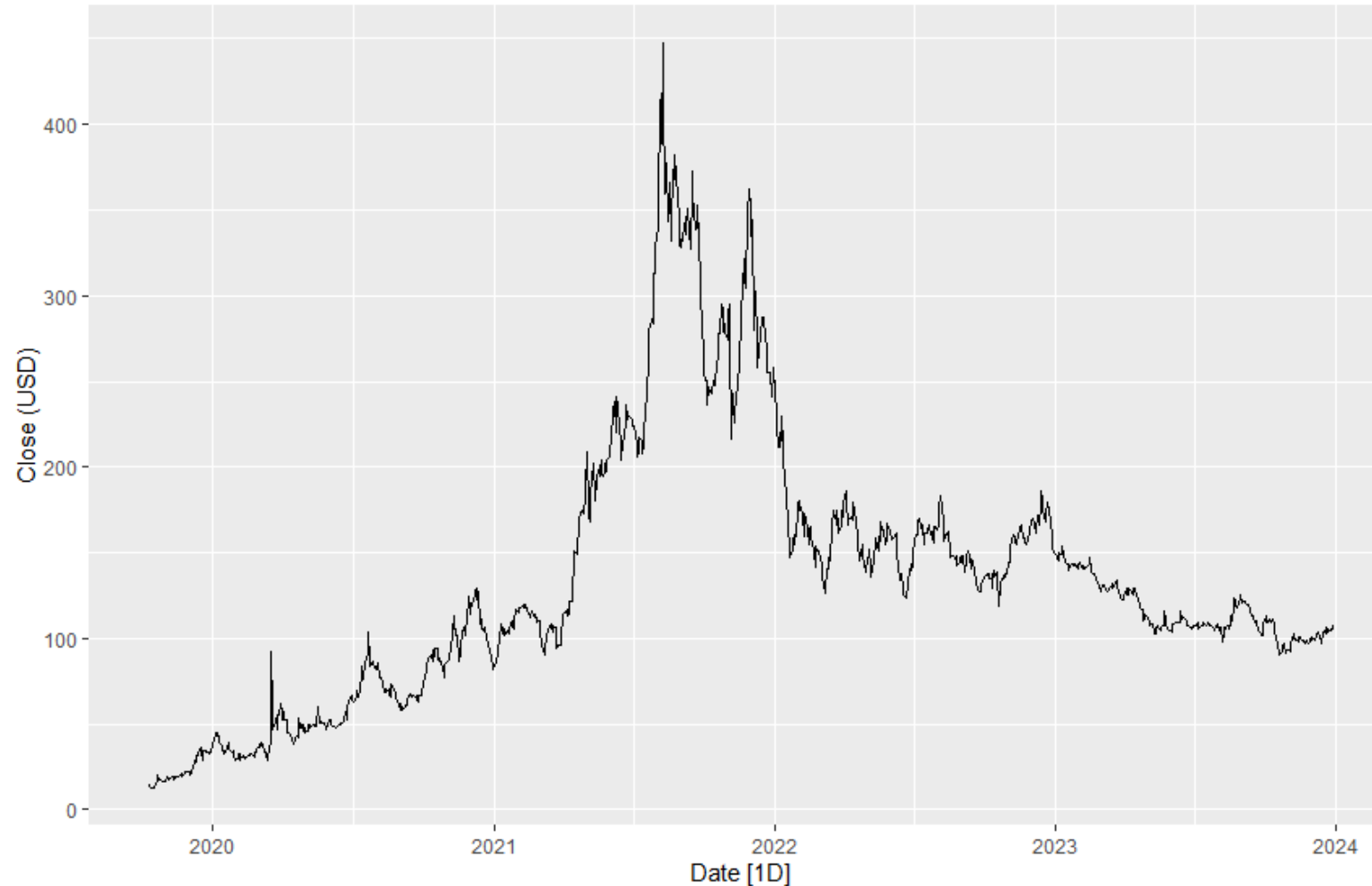
Shell (SHEL)

Daily closing price, 2019–2023



BioNTech (BNTX)

Daily closing price, 2019–2023



The *time series plots* can be used to identify initial structural breaks, which become particularly important later in the analysis.

Stationarity Tests

- ADF test: H_0 : time series is not stationary (unit root), H_1 : time series is stationary
- Stationary if tau value is smaller than critical value ($\alpha=5\%$)

Value of test-statistic is: **-1.2465** **2.6347**

Critical Values (ADF Test):

Test Statistic	1%	5%	10%
τ_2	-3.43	-2.86	-2.57
ϕ_1	6.43	4.59	3.78

→ Not stationary

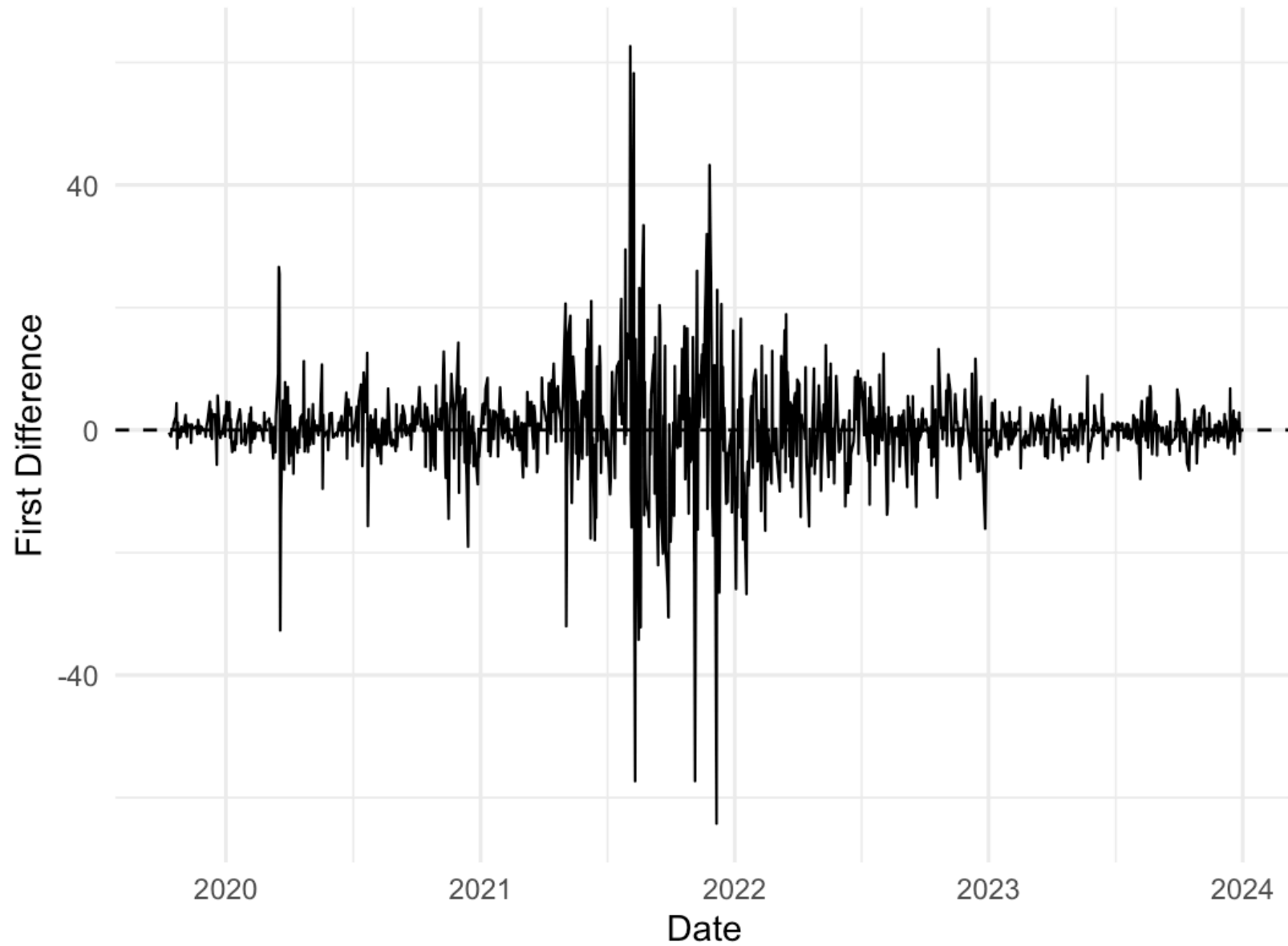
- $-1.2465 > -2.86$
- H_0 cannot be rejected

Time Series Becomes Stationary

- First differences: make the time series stationary
 - remove trend
 - changes in prices with stable variance around the mean
- provides reliable basis for future calculations

BioNTech (BNTX) – First Differences

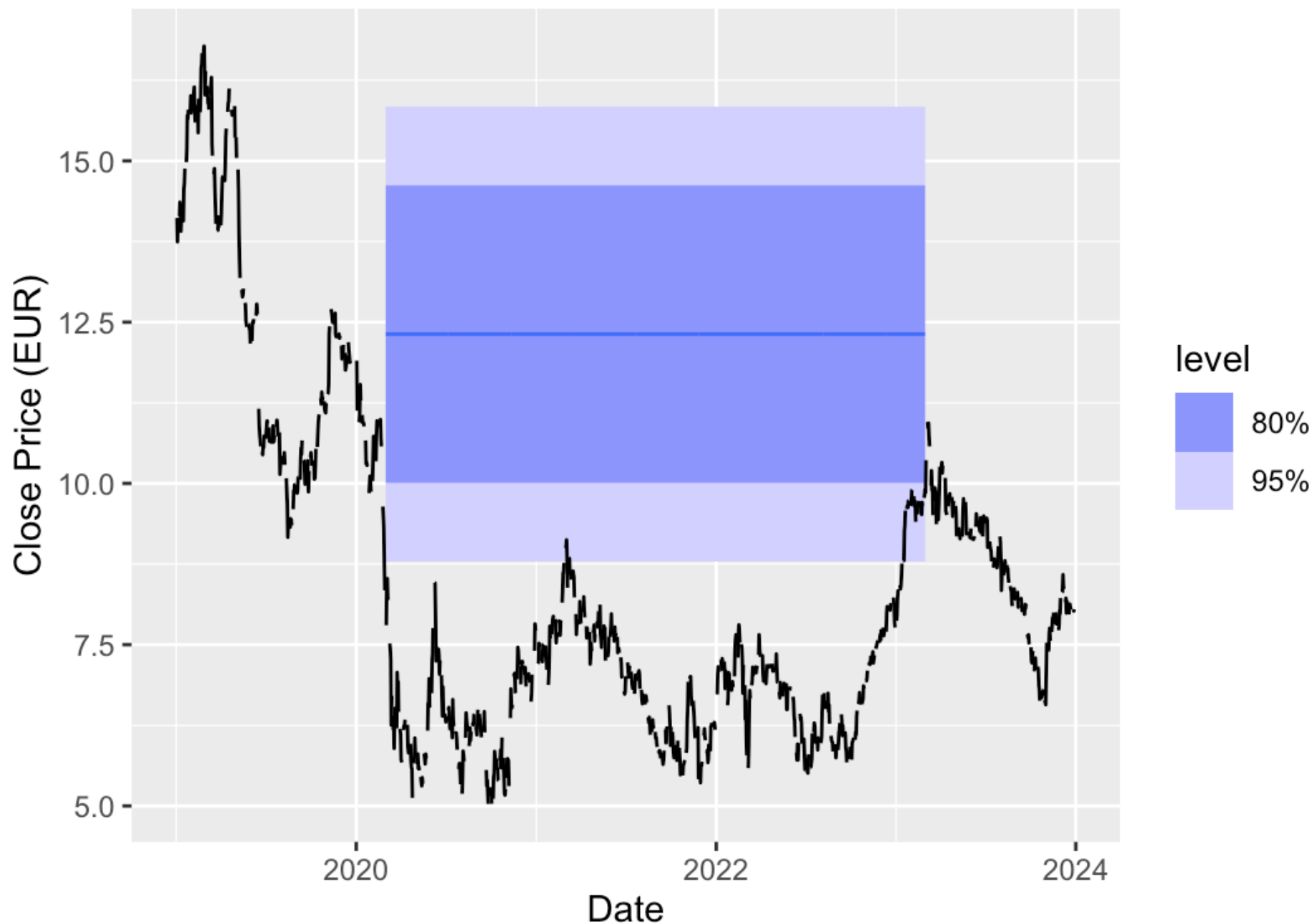
Daily price changes



Forecasting (Pre-COVID Benchmark)

Lufthansa (LHA.DE) — ARIMA Forecast vs Actual

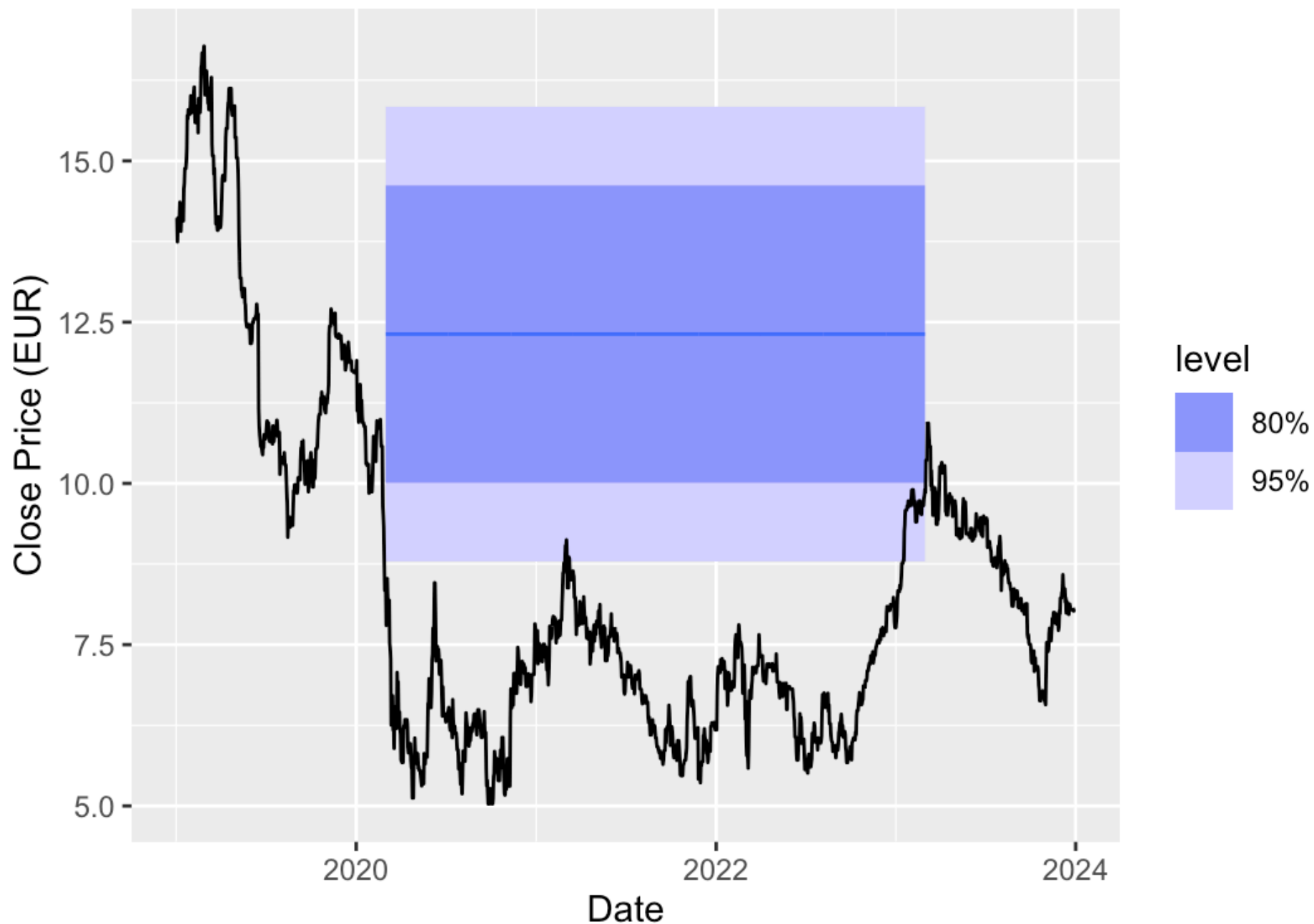
Model trained on data before March 2020



Continuous Forecast

Lufthansa (LHA.DE) — ARIMA Forecast vs Actual

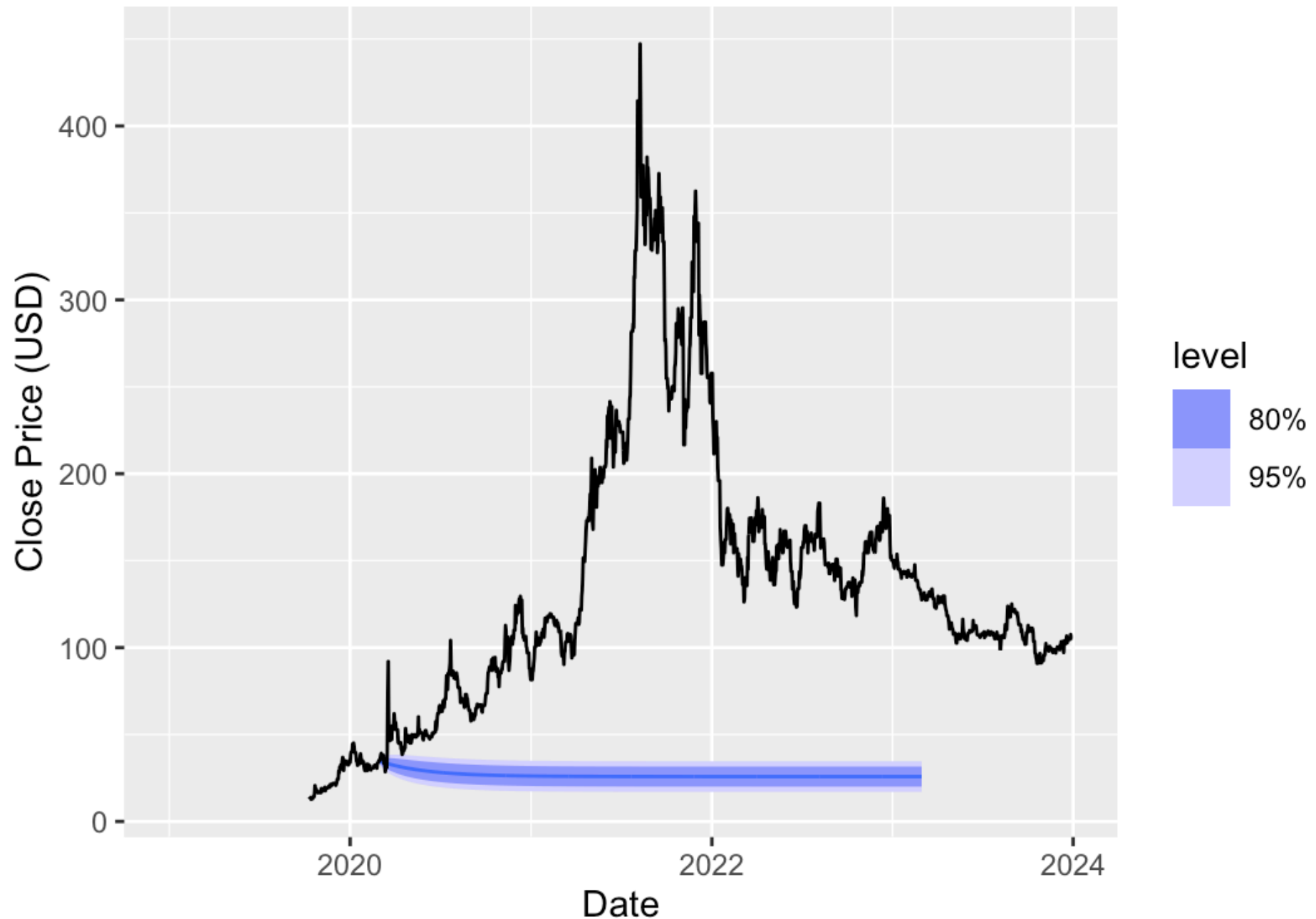
Model trained on data before March 2020





BioNTech (BNTX) — ARIMA Forecast vs Actual

Model trained on data before March 2020



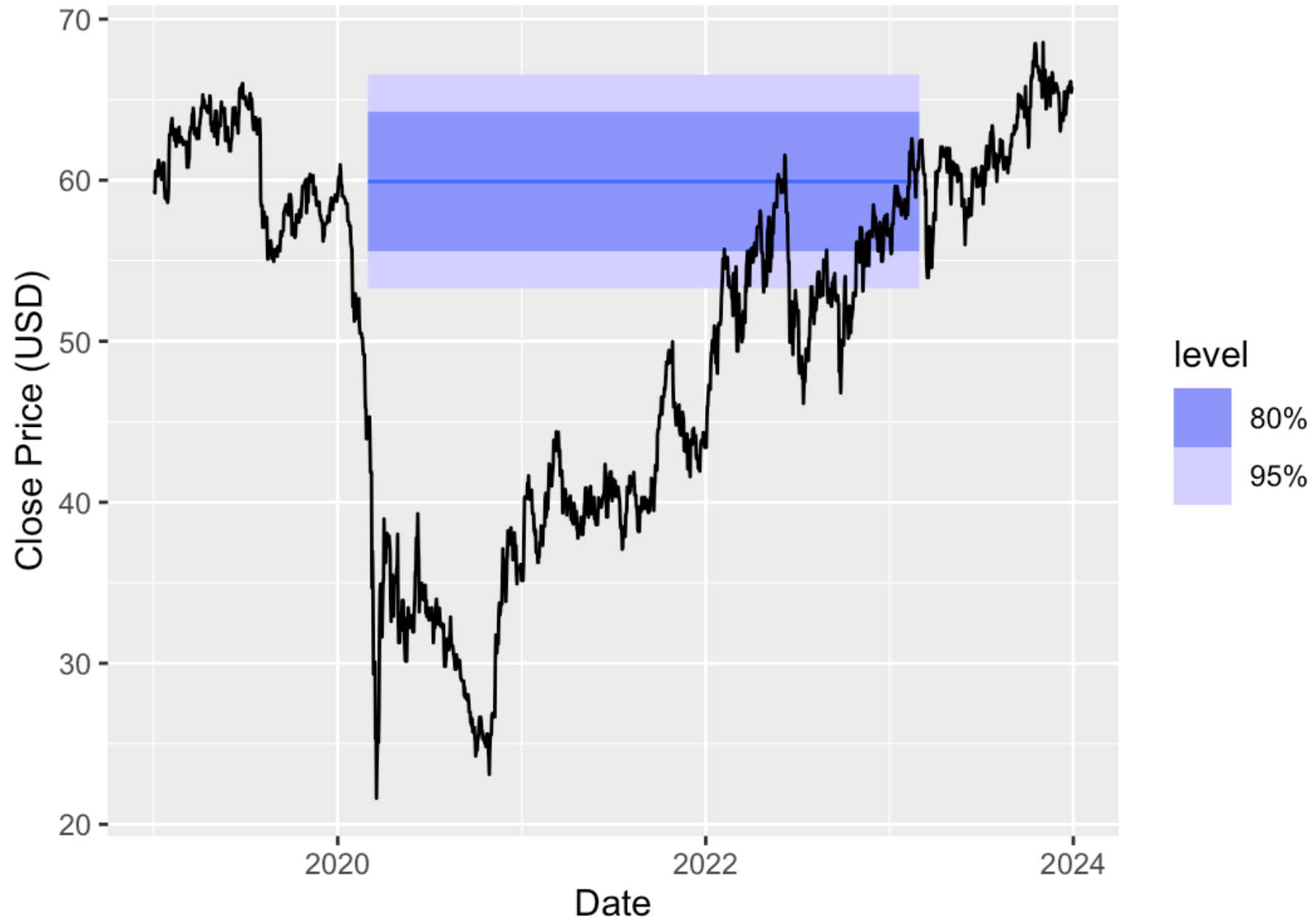
Black = actual price (visual forward fill only) Blue = ARIMA forecast

Forecast as a basis for determining recovery

- New idea: stock recovered when reaching 80% interval

Shell (SHEL) — ARIMA Forecast vs Actual

Model trained on data before March 2020



Black = actual price (visual forward-fill only), Blue = ARIMA forecast

Limitations so far

1. Stock Data set cleaning because of unallignment in rows
2. Stocks have different currencies → not as important at the moment, but could be a problem in the further process
3. COVID-Data not optional in original form as well → break down of data
4. Generally filtering the enormous amount of data according to our “needs”
5. Only considering stock prices in relation to confirmed Covid cases, also due to the limited time frame
6. Should we use the stationarity time series for the further procedure?

Conclusion

- Rejected the idea of the first question and settled on the current topic.
 - Found, extracted and cleaned of both Data Sets
 - Initial thought: divide the stock data set into quarterly segments
- Rejected because it would have simplified/misrepresented our results

- Summarized and compared the data collection in a compatible table
- Visualization of both data sets
- Tried on stationary tests
- Initial creation of forecasts based on the ARIMA model

Further Procedure

1. Test for long-run relationships (Cointegration)
2. Analyze short-run dynamics (Granger causality test)
3. Resulting in: possible sector-specific COVID–market linkages
4. Robustness checks with alternative specifications (monthly forecast eventually)

List of References

Stock price data obtained from Yahoo Finance via the `quantmod` package

Apple (AAPL): <https://finance.yahoo.com/quote/AAPL>

BioNTech (BNTX): <https://finance.yahoo.com/quote/BNTX>

Lufthansa (LHA.DE): <https://finance.yahoo.com/quote/LHA.DE>

Shell (SHEL): <https://finance.yahoo.com/quote/SHEL>

<https://github.com/CSSEGISandData/COVID-19>

Path of Confirmed cases (global):

`csse_covid_19_data/csse_covid_19_time_series/time_series_c`

Path of Deaths (global):

csse_covid_19_data/csse_covid_19_time_series/time_series_c

<https://otexts.com/fpp3/arma-reading.html>